a.Data[] b.Data[] a.Scale[] b.Scale[] c.Data

Data Unpacker

a.sign[i] b.sign[i] a.mant[i] b.mant[i] a.exp[i] b.exp[i]

Sign XOR

Man.MUL

HH 8x8 | HL 8x4 | LH 4x8 | LL 4x4

TF32FP16BF16INT8 FP4 | TF32FP16INT8FP4 | TF32FP16INT8FP8/6 FP4 | TF32FP16 FP8/6 FP4

Product Shift Add

Exp. ADD

8b Add 5b Add

Scale Unpacker

a.s.man.[i/g] b.s.man.[i/g] a.s.exp.[i/g] b.s.exp.[i/g]

MAX

U4 S8

Man.MUL Exp. ADD

U8 S9

inter_lane Anchor

THQ-Align Shifter | HQ-Align Shifter | Q-Align Shifter | Q-Align Shifter

28

Exp. Delta

FP4/INT8 path

Complement

FP4 Add | FP4 Add | FP4 Add | FP4 Add

29

Reduce ADD Tree

31

HiF4 shift

T1H2Q4E8

x N(2)

U8

Anchor Recomput

S10

ADD

32

UComplement

NV/HiF4 path

U14

Man.MUL

U8

Group Anchor

sign

SoP U31

LZD

zero_num U5

S10

Sub zn

S10

T2H4Q8E16

C Unpacker

x N(4)

C.mant[i]

Bi-Direct Shifter | Bi-Direct Shifter | Bi-Direct Shifter | Bi-Direct Shifter | Bi-Direct Shifter

INT8 path

INT32 {7b0}

39 39

Complement

s40

S10

C.exp

S11

Exp. Delta

inter_Group Anchor

S10

MAX

S10

CSA 4:2

CSA 3:2

ADD

s43

special

Anchor

INT PATH

LZD & Norm & Round

[38:7] s32

Float or Int

32 32

T8H16Q32E64